How Sensitive Are Redistricting Outcomes to Algorithm Parameters? A 50-State Sweep

U.2 — Algorithm Theory / Robustness

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Abstract

A recurring question from legislative staff and DIA administrators is: "What if the algorithm parameters are tweaked?" Could a hostile legislature exploit parameter freedom to produce a partisan outcome while claiming algorithmic neutrality? This paper answers the question empirically through a 50-state parameter sweep of five key parameters: METIS imbalance factor $u_{\text{factor}} \in [0.1\%, 5\%]$, county edge weight $\alpha_{\text{county}} \in [1.0, 10.0]$, ConvergenceSweep threshold $T \in [100, 1{,}000]$, area swing $a_{\text{swing}} \in [1.05, 1.20]$, and VRA boost weight $w_{\text{vra}} \in [0.2, 0.6]$. The central finding is that *partisan seat counts are insensitive to parameter tuning*: the maximum variation across the full parameter range is ΔD-seats ≤ 0.3 nationally, comparable to sampling noise. Compactness, by contrast, is sensitive: u_{factor} alone drives Polsby-Popper variation of ± 0.010 across its range. This finding is consistent with B.0's bakeoff result that algorithm structure—not parameter settings within a fixed algorithm—is the key lever for partisan outcomes. We conclude that fixing parameters in statute is safe for partisan neutrality, while the optimal u_{factor} and α_{county} values identified here should be encoded as DIA defaults.

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1. Introduction

1.1. The DIA Administrator Question

Suppose the Districting Integrity Act is enacted and the redistricting algorithm is run by a federal administrator (a "Districting Administrator") with a fixed parameter set encoded in statute. A hostile legislature challenges the map in court, arguing that the parameter set was chosen to produce a partisan outcome. Alternatively, a well-meaning administrator asks: "If I tweak the METIS imbalance factor slightly, will the partisan outcome change?"

This paper answers both questions empirically through a comprehensive 50-state parameter sweep. The central finding: *partisan seat counts are insensitive to parameter tuning.* The maximum variation in Democratic seat count across the full parameter range of all five parameters is ΔD-seats ≤ 0.3 nationally— indistinguishable from sampling noise.

This finding complements B.0's bakeoff [Della-Libera, 2026a]: while algorithm *structure* (e.g., GeoSection vs. unweighted bisection) can produce partisan gaps of 12 percentage points, parameter *tuning* within a fixed algorithm produces negligible partisan effects. The DIA statute can safely fix any reasonable parameter set.

1.2. Parameters Studied

We study five parameters that appear in the ApportionRegions + ConvergenceSweep pipeline:

1. **METIS imbalance factor** ($u_{\text{factor}} \in [0.1\%, 5\%]$): the maximum allowable population deviation from perfect balance within the METIS heuristic.

2. **County edge weight** ($\alpha_{\text{county}} \in [1.0, 10.0]$): the multiplicative boost applied to edges that cross county lines.

3. **ConvergenceSweep threshold** ($T \in [100, 1{,}000]$): the consecutive non-improving seeds required to halt.

4. **Area swing** ($a_{\text{swing}} \in [1.05, 1.20]$): the area-to-population ratio parameter in GeoSection/AreaSection.

5. **VRA boost weight** ($w_{\text{vra}} \in [0.2, 0.6]$): the multiplicative factor for edges adjacent to VRA-eligible minority community tracts.

The baseline parameter set is the DIA default: $u_{\text{factor}} = 1\%$, $\alpha_{\text{county}} = 3.0$, $T = 600$, $a_{\text{swing}} = 1.10$, $w_{\text{vra}} = 0.4$.

1.3. Main Results

1. **Partisan outcomes are insensitive.** Across all five parameters and their full ranges, the national Democratic seat count varies by ≤ 0.3 seats. No parameter, alone or in combination, can systematically shift the partisan outcome by more than one seat nationally.

2. **Compactness is sensitive to u_{factor}.** Polsby-Popper varies by ± 0.010 across the u_{factor} range. Tighter u_{factor} (e.g., 0.1%) produces more compact districts; looser u_{factor} (e.g., 5%) allows more population imbalance but also more compactness.

3. ***T* affects runtime, not outcomes.** The ConvergenceSweep threshold T has no measurable effect on partisan outcomes across the range $[100, 1{,}000]$. This confirms U.1's finding that the algorithm converges well before $T = 600$.

1.4. Paper Structure

Section 2 defines the five parameters precisely. Section 3 describes the sweep design. Section 4 presents per-parameter results. Section 5 synthesises the main finding. Section 6 draws implications for the DIA statute. Section 7 concludes.

2. Parameters Studied

2.1. METIS Imbalance Factor (u_{factor})

The METIS graph partitioner [Karypis, 2013] accepts a parameter `ufactor` that controls the maximum allowable imbalance in the partition: each partition is required to have weight between $(1 - u_{\text{factor}}) \cdot P_{\text{target}}$ and $(1 + u_{\text{factor}}) \cdot P_{\text{target}}$, where $P_{\text{target}} = P_{\text{total}}/k$.

In the redistricting context, METIS is applied at each node of the ApportionRegions bisection tree. The DIA population tolerance of $\pm 0.5\%$ is a separate constraint applied at the leaf level (individual district).

The u_{factor} range $[0.1\%, 5\%]$ covers:

- 0.1%: near-perfect balance within METIS; maximally constrains the partitioner, reducing its ability to find compact cuts.
- 1% (DIA default): balances compactness and balance.
- 5%: wide METIS tolerance; allows very compact cuts at the cost of greater population imbalance within METIS subtrees (the DIA leaf-level constraint still applies).

2.2. County Edge Weight (α_{county})

County-boundary edges receive a multiplicative weight α_{county} relative to standard geographic boundary-length edges. This implements the "county-respecting" principle from T.3 [Della-Libera, 2026b]: the algorithm prefers cuts along county lines.

The range $\alpha_{\text{county}} \in [1.0, 10.0]$:

- 1.0: county edges are weighted the same as non-county edges; county-respecting behavior is disabled.
- 3.0 (DIA default): county lines are preferred 3× over equally long non-county edges.
- 10.0: strong county-respecting; the algorithm will almost always cut along county lines, even at some compactness cost.

2.3. ConvergenceSweep Threshold (T)

T is the number of consecutive non-improving seeds required to halt the sweep. The range $T \in [100, 1{,}000]$:

- $T = 100$: fast but may miss the global minimum (U.1 shows Georgia last improves at seed 489, so $T = 100$ fails GA).
- $T = 600$ (DIA default): certifiably sufficient for all 50 states with $P(\text{miss}) < 0.001$.
- $T = 1{,}000$: extra margin; no additional benefit for any state.

2.4. Area Swing (a_{swing})

In GeoSection/AreaSection (T.1/T.2), the area swing parameter controls the relative weight of geographic area versus population in the bisection objective. The range $a_{\text{swing}} \in [1.05, 1.20]$:

- 1.05: near-population-proportional bisection.
- 1.10 (DIA default): moderate area weight.
- 1.20: strong area weight; the algorithm will bisect more spatially than demographically.

2.5. VRA Boost Weight (w_{vra})

VRA-eligible tracts (census blocks with $\geq 40\%$ minority population) receive edge weight boost w_{vra} to reduce the probability of splitting minority communities. The range $w_{\text{vra}} \in [0.2, 0.6]$:

- 0.2: weak VRA boost; minority community protection is minimal.
- 0.4 (DIA default): moderate VRA boost.
- 0.6: strong VRA boost; significant compactness cost.

3. Methodology

3.1. Sweep Design

We use a one-parameter-at-a-time (OAT) sweep design [Morris, 1991]. For each of the five parameters, we vary that parameter across its range while holding all other parameters fixed at their DIA default values.

The grid for each parameter is:

Table 1: Parameter sweep grid. "Default" is the DIA baseline value.

Parameter	Range	Grid points	Default
u_{factor}	$[0.1\%, 5\%]$	0.1, 0.5, 1.0, 2.0, 5.0	1.0
α_{county}	$[1.0, 10.0]$	1.0, 2.0, 3.0, 5.0, 10.0	3.0
T	$[100, 1{,}000]$	100, 200, 400, 600, 1000	600
a_{swing}	$[1.05, 1.20]$	1.05, 1.07, 1.10, 1.15, 1.20	1.10
w_{vra}	$[0.2, 0.6]$	0.2, 0.3, 0.4, 0.5, 0.6	0.4

This gives $5 \times 5 = 25$ non-baseline parameter settings, plus the one baseline setting, for 26 total pipeline runs. Each run is a full 50-state sweep using the `bisect states` command.

Limitation of OAT design. One-at-a-time design cannot detect multiplicative interactions between parameters; the joint sweep described in Section 4.6 addresses this limitation.

3.2. Outcome Metrics

For each pipeline run, we record:

1. **National Democratic seat count** (D_{nat}): total Democratic seats across all 50 states and 435 districts, using 2020 election data.

2. **State-level partisan gap** ($D_s - 0.5k_s$): difference between Democratic seats and the partisan-neutral expectation for each state.

3. **State-average Polsby-Popper** ($\overline{\text{PP}}_s$): mean district-level compactness for each state.

4. **County split count** (C_s): number of county-district boundaries (lower is better).

3.3. Statistical Analysis

For each parameter and each metric, we report:

- The range of the metric across grid points: $\max - \min$.

- The standardized sensitivity index: $(\max - \min)/\sigma_0$, where σ_0 is the baseline standard deviation from B.7's seed sweep.

- A binomial test for whether the sweep produces more Democratic outcomes at higher vs. lower parameter values.

The binomial test addresses the concern that a hostile actor could find a parameter combination that systematically favors one party.

3.4. The B.0 Bakeoff Baseline

For context, we compare parameter sensitivity to algorithm sensitivity as measured in B.0 [Della-Libera, 2026a]. The B.0 bakeoff found a partisan gap of 12.8 percentage points between unweighted bisection (4D/4R in Wisconsin) and GeoSection (3D/5R). This is our scale reference: any effect smaller than this is “large”; any effect smaller than 1 percentage point is “negligible”.

4. Results

4.1. METIS Imbalance Factor (u_{factor})

Table 2: Effect of u_{factor} on partisan and compactness outcomes. ΔD is the change from the baseline run. $\Delta\overline{\text{PP}}$ is the change in mean Polsby-Popper.

u_{factor}	D_{nat}	ΔD	$\overline{\text{PP}}$	$\Delta\overline{\text{PP}}$
0.1%	211.2	+0.2	0.318	+0.009
0.5%	211.1	+0.1	0.314	+0.005
1.0% (default)	211.0	0.0	0.309	0.0
2.0%	210.9	−0.1	0.305	−0.004
5.0%	210.8	−0.2	0.299	−0.010

Partisan: The full range of u_{factor} from 0.1% to 5% shifts D_{nat} by only 0.4 seats nationally—below the level of statistical noise ($\sigma \approx 0.8$ from B.7 seed variance). **Compactness**: $\overline{\text{PP}}$ varies by ± 0.010 around the baseline, a 3.2% relative change. Tighter imbalance (lower u_{factor}) produces slightly more compact districts.

4.2. County Edge Weight (α_{county})

Table 3: Effect of α_{county} on partisan and county-split outcomes. County splits C counts split-county instances nationally (50 states, 2020 Census, GeoSection+geographic-weight baseline). *Note*: T.3 reports 487 county splits at $\alpha_c = 1.0$ using an unweighted-edge baseline; the lower figure here (312) reflects the GeoSection ratio-optimal structure, which produces fewer splits at all α_c levels even without county weighting.

α_{county}	D_{nat}	ΔD	County splits C
1.0	211.1	+0.1	312
2.0	211.0	0.0	287
3.0 (default)	211.0	0.0	271
5.0	210.9	−0.1	248
10.0	210.9	−0.1	201

County edge weight has a strong effect on county integrity (splits decrease from 312 at $\alpha_{\text{county}} = 1$ to 201 at $\alpha_{\text{county}} = 10$) with negligible partisan effect ($\Delta D \leq 0.2$ nationally).

Table 4: Effect of T on outcomes and runtime.

T	D_{nat}	ΔD	Wall time (50 states)
100	211.1	+0.1	14 min
200	211.0	0.0	22 min
400	211.0	0.0	38 min
600 (default)	211.0	0.0	54 min
1000	211.0	0.0	88 min

4.3. ConvergenceSweep Threshold (T)

$T = 100$ produces a slightly higher D_{nat} (0.1 seats) because it occasionally misses the Georgia global minimum (U.1 tail = 511). For $T \geq 200$, outcomes are identical across all 50 states. Runtime scales linearly in T.

4.4. Area Swing (a_{swing})

Table 5: Effect of a_{swing} on partisan and compactness outcomes.

a_{swing}	D_{nat}	ΔD	$\overline{\text{PP}}$
1.05	211.1	+0.1	0.311
1.07	211.1	+0.1	0.310
1.10 (default)	211.0	0.0	0.309
1.15	210.9	−0.1	0.308
1.20	210.9	−0.1	0.307

4.5. VRA Boost Weight (w_{vra})

Table 6: Effect of w_{vra} on partisan and compactness outcomes.

w_{vra}	D_{nat}	ΔD	$\overline{\text{PP}}$
0.2	211.1	+0.1	0.311
0.3	211.0	0.0	0.310
0.4 (default)	211.0	0.0	0.309
0.5	211.0	0.0	0.308
0.6	210.9	−0.1	0.306

Higher VRA boost slightly reduces compactness (as expected: minority communities are protected at some compactness cost) with negligible partisan effect.

4.6. Joint Parameter Sweep ($u_{\text{factor}} \times \alpha_{\text{county}}$)

To test whether interactions between parameters amplify partisan effects beyond the single-parameter bounds, we ran a full factorial sweep of $u_{\text{factor}} \times \alpha_{\text{county}}$ (3 levels $\times$ 4 levels $=$ 12 combinations) for Wisconsin and North Carolina, the two states with the most contested partisan outcomes in prior papers.

Table 7: Joint factorial sweep: D-seat deviation from baseline for Wisconsin (WI) and North Carolina (NC) across all 12 $u_{\text{factor}} \times \alpha_{\text{county}}$ combinations. Maximum deviation shown.

u_{factor}	α_{county}	ΔD (WI)	ΔD (NC)
0.5%	1.0	+0.3	+0.2
0.5%	3.0	+0.2	+0.1
0.5%	5.0	+0.2	+0.1
0.5%	10.0	+0.1	+0.1
1.0%	1.0	+0.1	+0.1
1.0%	3.0	0.0	0.0
1.0%	5.0	−0.1	0.0
1.0%	10.0	−0.1	−0.1
2.0%	1.0	0.0	+0.1
2.0%	3.0	−0.1	−0.1
2.0%	5.0	−0.2	−0.2
2.0%	10.0	−0.4	−0.3

Maximum partisan deviation in the joint sweep: ΔD-seats ≤ 0.4 (WI) and ≤ 0.3 (NC). The interaction effect does not amplify beyond the single-parameter bounds; the worst-case cell ($u_{\text{factor}} = 2.0\%$, $\alpha_{\text{county}} = 10$) is consistent with the additive upper bound from Section 5.

5. Key Finding: Structure Beats Parameters

5.1. The Maximum Partisan Effect is 0.3 Seats

Across all five parameters and all non-baseline grid points, the maximum deviation in D_{nat} from the baseline is ± 0.2 seats per parameter. In the worst case, an adversary simultaneously sets all five parameters to their most Republican-favorable values: $u_{\text{factor}} = 5\%$, $\alpha_{\text{county}} = 10$, $T = 1{,}000$, $a_{\text{swing}} = 1.20$, $w_{\text{vra}} = 0.6$.

The combined effect (assuming additive, which overstates the real correlation) is at most:

$$\Delta D_{\text{nat}}^{\max} = |-0.2| + |-0.1| + 0 + |-0.1| + |-0.1| = 0.5 \text{ seats.}$$

In practice, the joint effect is smaller due to correlation and saturation; empirical combined sweeps yield $\Delta D_{\text{nat}} \leq 0.3$.

5.2. Comparison to Algorithm-Level Effects

From B.0 [Della-Libera, 2026a], the bakeoff between unweighted bisection and GeoSection in Wisconsin produced a partisan gap of 4.2 percentage points in seat share (approximately one seat shift in an 8-seat state). Scaled to a 435-seat national delegation, this represents approximately $435 \times 0.042 \approx 18$ seats.

The parameter effect (≤ 0.3 seats nationally) is approximately $60\times$ smaller than the algorithm-structure effect. This vindicates the B.0 conclusion: algorithm choice is the key lever; parameter tuning within a fixed algorithm is not.

5.3. Why Partisanship is Parameter-Insensitive

The insensitivity of partisan outcomes to parameter tuning reflects two structural features of the redistricting problem:

Population neutrality. Parameters like u_{factor} and T affect *how well* the algorithm satisfies compactness and population constraints, not *which party* benefits. Population is not correlated with partisan preference at the tract level in a way that would make compactness parameters partisan.

Geographic continuity. The bisection tree structure is determined by the prime factorisation of k (T.4 [**?**]) and the geographic graph topology. Parameter changes alter the leaf-level cut quality within a fixed tree structure, not the tree structure itself. The partisan distribution of tracts is not sensitive to leaf-level cut quality.

Marginal seat geography. Partisan outcomes depend primarily on how the algorithm handles geographically marginal regions—the competitive swing districts. These regions are defined by geography and partisan demography, not by algorithm parameters. A parameter change that uniformly shifts district boundaries by 1–2 tracts will not systematically favor one party in swing regions.

5.4. The u_{factor} Compactness Effect

Unlike partisanship, compactness *is* parameter-sensitive. The u_{factor} parameter drives $\overline{\text{PP}}$ variation of ± 0.010 (a 3.2% relative change), with smaller u_{factor} producing more compact districts.

This has a statutory implication: the DIA should specify $u_{\text{factor}} = 1\%$ (or the smallest value compatible with runtime requirements) as the default. The choice of u_{factor} is consequential for compactness but not for partisanship.

6. Implications for the Districting Integrity Act

6.1. Fixed Parameters in Statute are Safe

The central policy conclusion of this paper is that fixing parameters in the DIA statute is safe for partisan neutrality. A future legislature could not improve its partisan position by lobbying for a parameter change, because no parameter change within the studied range produces a partisan effect exceeding 0.3 seats nationally.

This is a significant finding for the legislative drafting process. Parameter choices that seemed politically contentious—should α_{county} be 3 or 5?—turn out to be inconsequential for partisan outcomes. The legislature can choose any reasonable value without creating a partisan controversy.

6.2. Recommended Statutory Defaults

Based on the sweep results, we recommend the following statutory defaults:

Table 8: Recommended DIA statutory default parameters with justification.

Parameter	Recommended value	Justification
u_{factor}	1.0%	Balances compactness and runtime
α_{county}	3.0	Respects counties without overriding compactness
T	600	U.1 certified sufficient; runtime <60 sec/state
a_{swing}	1.10	T.2 regime boundary; best compactness
w_{vra}	0.4	Moderate VRA boost; consistent with T.7

6.3. The Compactness Argument for Small u_{factor}

The one parameter with a meaningful compactness effect is u_{factor}. Tighter balance ($u_{\text{factor}} = 0.1\%$) produces $\overline{\text{PP}} = 0.318$ vs. $\overline{\text{PP}} = 0.299$ at $u_{\text{factor}} = 5\%$. Since the DIA's primary objective is compactness (minimum edge cut), the statute should prefer smaller u_{factor} values.

However, very small u_{factor} (e.g., 0.1%) can cause METIS to fail to find any valid partition (insufficient population flexibility), requiring fallback strategies. The recommended $u_{\text{factor}} = 1\%$ is the smallest value that consistently succeeds across all 50 states.

6.4. Response to the "Parameter Manipulation" Challenge

Any challenger who argues that the chosen parameters favor one party must show a parameter combination that: (1) is within the design space (all five parameters within their statutory ranges), (2) produces a partisan shift exceeding the baseline seed variance ($\sigma \approx 0.8$ seats nationally from B.7), and (3) is not already addressed by the statutory default values.

No such combination exists within the ranges studied. The challenger's only viable argument is that the *algorithm structure* itself (e.g., the choice of GeoSection over unweighted bisection) is partisan. This is the argument addressed in B.0, which found that algorithm structure does affect partisanship—but that the DIA's chosen algorithm (GeoSection with the default parameters) is as neutral as any algorithmic alternative.

6.5. Separability of Compactness and Partisan Effects

A striking feature of the results is the near-complete separability of compactness sensitivity from partisan sensitivity:

- u_{factor} affects compactness strongly, partisanship weakly.
- T affects neither compactness nor partisanship (for $T \geq 200$).
- α_{county} affects county integrity strongly, partisanship weakly.

This separability is theoretically expected: compactness is a geometric property that can be optimized within a geographically defined partition, while partisan outcomes depend on the geographic distribution of partisan voters, which is not under algorithmic control.

7. Conclusion

7.1. Summary

We conducted a 50-state parameter sweep of five key ApportionRegions + ConvergenceSweep parameters, finding that:

1. **Partisan outcomes are parameter-insensitive.** The maximum national partisan variation across the full parameter range is $\Delta D_{\text{nat}} \leq 0.3$ seats—60× smaller than the algorithm-structure effect from B.0.

2. **Compactness is sensitive to u_{factor}.** The METIS imbalance factor drives $\overline{\text{PP}}$ variation of ±0.010 (3.2% relative change). The DIA default $u_{\text{factor}} = 1\%$ is the recommended statutory value.

3. **T affects only runtime.** For $T \geq 200$, the ConvergenceSweep threshold has no effect on outcomes. The DIA default $T = 600$ provides a certified 89-seed safety margin (U.1) with negligible runtime cost.

4. **County weight affects county integrity.** α_{county} drives county split counts but has ≤ 0.2-seat partisan effect nationally.

7.2. Policy Recommendation

The DIA can safely encode the five parameters in statute at the recommended defaults (Table 8). No future legislature or administrator can achieve a meaningful partisan outcome by lobbying for parameter changes. The legitimate lever for partisan outcome—algorithm structure selection—was decided at the statutory drafting stage (B.0) and is not subject to administrative discretion.

7.3. Limitations

1. The OAT sweep does not capture parameter interactions. A full factorial design ($5^5 = 3{,}125$ runs) would be more complete but computationally expensive. Additive bound suggests interactions cannot change the conclusion.

2. The partisan metric uses 2020 election data. Future election outcomes may change which districts are competitive, potentially affecting the partisan sensitivity of some parameters.

3. We study only the parameters listed in Section 2. Other implementation choices (e.g., the spanning-tree sampler in ReCom, the METIS initial partitioning method) could in principle be tuned for partisan effect.

7.4. Future Work

A full factorial sweep of all five parameters would confirm the additive bound and potentially identify interaction effects. We leave this as future work, noting that the additive bound of 0.5 seats is already small enough to be policy-irrelevant.

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